FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
FULL TEST – VII
PAPER –1
TEST DATE: 07-08-2022

ANSWERS, HINTS & SOLUTIONS

Physics PART – I

Section – A

1. A

Sol. $i_1 = \frac{\varepsilon}{R}e^{-\frac{t}{RC}}$

$i_2 = -\frac{2\varepsilon}{R}e^{-\frac{t}{RC}}$

$i_3 = -\frac{\varepsilon}{R}e^{-\frac{t}{RC}}$

2. C

Sol. Frequencey received by approaching car

$f_1 = f_0\left[1+\frac{2}{330}\right]$

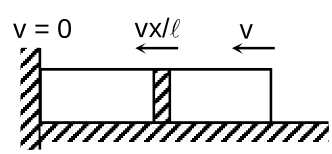
Frequency received by source again

$f_2 = \frac{f_1}{\left(1-\frac{2}{330}\right)}$

So, beat frequency $f_B = f_2 - f_0 = 6$ Hz.

3. A

Sol. At any moment total energy of rod



E = K + U

$= \int dk + \frac{1}{2}\frac{AY}{\ell}x^2$

$= \frac{1}{6}mv^2 + \frac{1}{2}\frac{AY}{\ell}x^2$

$\Rightarrow \frac{dE}{dt} = \frac{ma}{3} + \frac{AY}{\ell}x = 0$

FIITJEE *Ltd.*, *FIITJEE House, 29-A, Kalu Sarai, Sarvapriya Vihar, New Delhi -110016, Ph 46106000, 26569493, Fax 26513942*
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$\Rightarrow \omega = \sqrt{\dfrac{3AY}{\ell m}}$

4. B

Sol. Similar to gravitational force and orbital motion

$\Rightarrow v_{escape} = \sqrt{2}v_{orbital}$

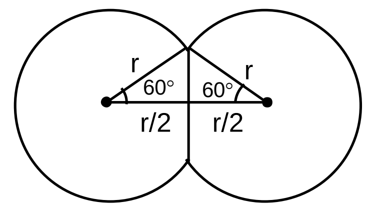
5. A, B

Sol. Deviation at h height = A(μ –1)

$= \dfrac{Ah}{k} = \dfrac{h}{k/A}$ similar to converging lens

6. B, C

Sol. In this case due to symmetry common surface will be flat and centers will be at r separation.



Lost surface area = $2\left[2\pi r^2 - \pi\dfrac{3}{4}r^2\right] = \dfrac{5}{2}\pi r^2$

Lost energy = $S\cdot\dfrac{5}{2}\pi r^2$

7. B

Sol. $P_{in} = \dfrac{4S}{R} + \dfrac{4S}{2R} = \dfrac{6S}{R}$

$P_{mid} = \dfrac{4S}{2R} = \dfrac{2S}{R}$

as T = constant

$\dfrac{P_{in}}{P_{mid}}\dfrac{V_{in}}{V_{mid}} = \dfrac{\mu_{in}}{\mu_{mid}} = \dfrac{3}{7} = y$

and $\dfrac{P_{in}}{P_{mid}} = \dfrac{\rho_{in}}{\rho_{mid}} = 3 = x$

8. A, C

Sol. Cylinder absorbs energy from left face and radiate from both, So

$\dfrac{P}{2}(1-\cos 60°) = \sigma AT^4 + \sigma AT'^4$

$\Rightarrow$ P = 68 W

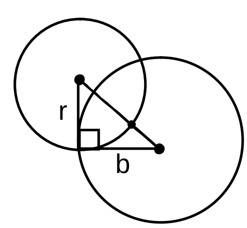
Heat current = 1 W

$\dfrac{kA\Delta T}{\ell} = 1$

$\Rightarrow$ k = 0.057

9. A, B

Sol. $\delta = \pi - 2\tan^{-1}\left(\dfrac{r}{b}\right) = \pi - 2\tan^{-1}\left(\dfrac{mv}{qBb}\right)$



If n > 1 $\Rightarrow \pi - 2\tan^{-1}x = \sin^{-1}\left(\dfrac{2x}{1+x^2}\right) = \sin^{-1}\left(\dfrac{2}{x}\right) = \dfrac{2}{x}$

$\delta = \dfrac{2qBb}{mv}$

10. A, D

Sol. $Bv\ell = L\frac{di}{dt} + iR \Rightarrow Bs\ell = Li + qR$

$Bi\ell = ma \Rightarrow Bq\ell = m(v_0 - v)$

$v = v_0 - \frac{B\ell}{Rm}(Bs\ell - Li)$

$\frac{B^2\ell^2}{mR} = \alpha$, $\frac{B\ell L}{mR} = \beta$

Section – B

11. 98

Sol. $V_P = \varepsilon e^{-t\frac{2R}{L}}$

$V_Q = \varepsilon\left(1 - e^{-\frac{tR}{L}}\right)$

$V_P - V_Q = 0$

$\Rightarrow e^{-\frac{2tR}{L}} + e^{-\frac{tR}{L}} - 1 = 0$

$e^{-\frac{tR}{L}} = \frac{-1+\sqrt{1+4}}{2} = \frac{\sqrt{5}-1}{2}$

12. 2

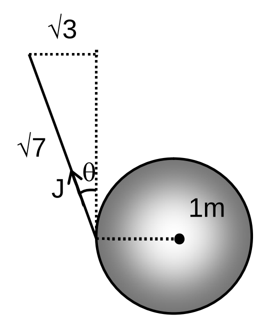
Sol. Impulse equation

$J\cos\theta = m\sqrt{2gR}$

Angular impulse equation

$J\cos\theta R = \frac{mR^2}{2}\omega$

$\omega = 2\sqrt{20}$



13. 4

Sol. PA = kx and V = Ax $\Rightarrow$ PV = kx^2 $\Rightarrow$ nRT = kx^2

dQ = nC_VdT + PdV

$= \frac{3}{2}nRdT + PdV$

= 3kx dx + kx dx = 4kx dx

Work = PdV = kxdx

Section – C

14. 00000.50

15. 00002.50

Sol. (for Q. 14-15)

In steady state photo current = $\frac{IAe}{hf} = \frac{V}{R}$

16. 00000.52

17. 00000.58

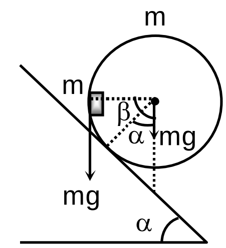
Sol. (for Q. 16-17)

Torque equilibrium about contact point

mg R sin $\alpha \leq$ mg(R sin β – R sin α)

$\Rightarrow$ 2 sin $\alpha \leq$ sin β

2 sin $\alpha \leq 1 \Rightarrow \alpha = 30°$



18. 00000.11

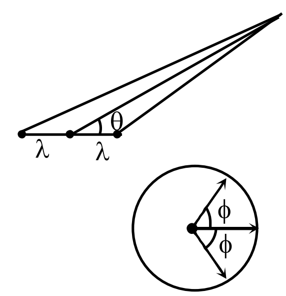
19. 00005.83

Sol. (for Q. 18-19)

Path difference of S_1 and S_2 with S

$\Delta x = \lambda \cos \theta$

Resultant amplitude = A + 2A cos ($2\pi \cos \theta$)



Chemistry

PART – II

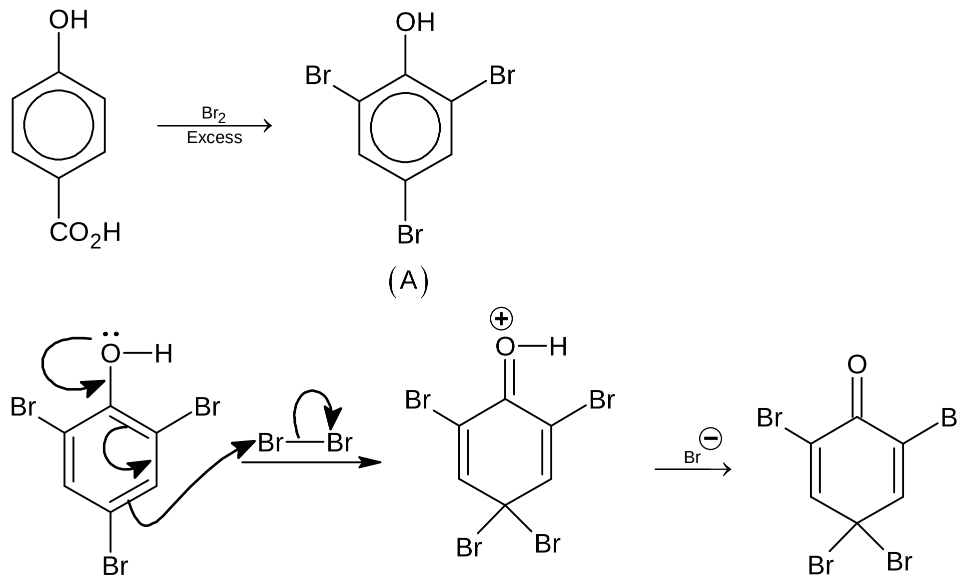
Section – A

20. B

Sol. $2\pi r = n\lambda$

21. C

Sol.



22. C

Sol. $Ag_3PO_4(s) \rightleftharpoons 3Ag^+(aq) + PO_4^{3-}(aq)$

$K_{SP} = 27\ S^4 = 2.7\times10^{-15}$

$= 27\times10^{-16}$

$S = 10^{-4}$

Total ions $= 4\times S = 4\times10^{-4}$

Relative lowering of vapour pressure $= \dfrac{4\times10^{-4}\times18}{1000} = 7.2\times10^{-6}$

% decrement of vapour pressure $= 7.2\times10^{-4}$.

23. A

Sol. $A(g) \rightleftharpoons 2B(g);\ K_P = \dfrac{2^2}{4} = 1$ atm

At new equilibrium, let partial pressure of A(g) = x atm, and partial pressure of B (g) = y atm.
Then, total pressure = (x + y) = 12 atm.

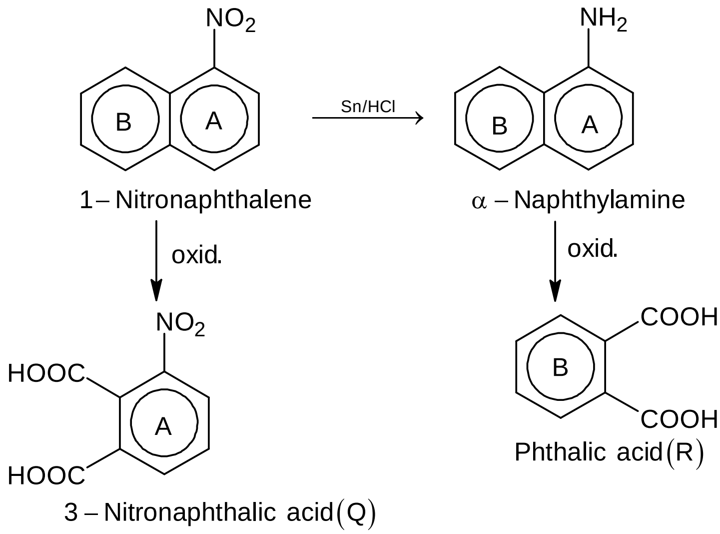
$K_P = \dfrac{y^2}{x} = 1$; ; solving for x and y, we get partial pressure of B (g) = 3 atm.

24. A, B, C

Sol. Stereospecific syn addition takes place in case of (A), (B) and (C). (D) is the reagent for hydration (HBO) not hydrogenation.

25. B, C

Sol.



The electron-attracting $-NO_2$ stabilizes ring A of 1-nitronaphthalene to oxidation, and ring B is oxidized to form 3-nitrophthalic acid. By orbital overlap, $-\ddot{N}H_2$ releases electron density, making ring A more susceptible to oxidation, and $\alpha-$naphthylamine is oxidized to phthalic acid. The NO_2 labels one ring and establishes the presence of two fused benzene rings in naphthalene.

26. C, D

Sol. All biodegradable polymers are not natural polymers.

27. A, B, C

Sol. In liquid or solid form Cl_2O_6 ionizes into chloryl perchlorate, $[ClO_2]^+[ClO_4]^-$ ClO_2^+ is planar.

$HNO_2 \xrightarrow{\Delta} NO_2 + NO + H_2O$

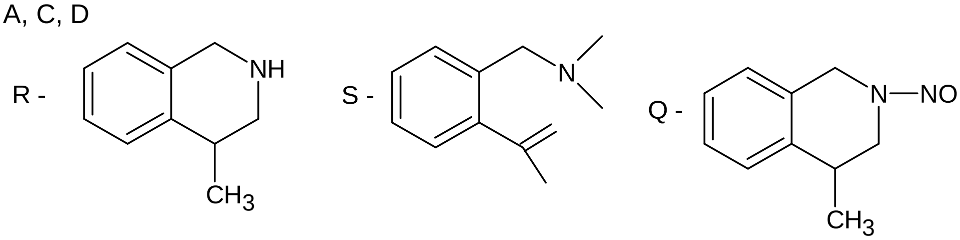
$3HClO_3 \xrightarrow{\Delta} HClO_4 + Cl_2 + 2O_2 + H_2O$

28. A, B

Sol. ${}^{40}_{19}K + e^- \longrightarrow {}^{40}_{18}Ar$; isodiaphers having same A – 2Z, Po belongs to group 16.

29. A, C, D

Sol.



Yellow oily N-nitrosoamine

Section – B

30. 3

Sol. $FC = V - N - \frac{B}{2}$; FC = formal charge; V = number of valence electrons.

N = number of nonbonding valence electrons. B = Total number of electrons shared in bonds

31. 1

Sol. 2 milimoles of acid weighs 0.2 g

Molar mass of acid = 100

$\Delta T_f = i \times k_f \times m;\ i = 1.5$

$$0.279 = 1.5 \times 1.86 \times \frac{x}{0.1 \times 100}$$

$$x = \frac{2.79}{1.5 \times 1.86}$$

= 1

32. 6

Sol.

$$H-\underset{\underset{O}{\parallel}}{C}-H \xrightarrow{\text{Conc. NaOH}} H-\overset{\overset{O}{\parallel}}{C}-O^{-}Na^{+} + CH_3OH$$

It is a disproportionation reaction

$$E = \frac{M}{2} + \frac{M}{2} = M = 30$$

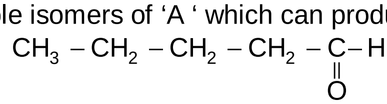
Section – C

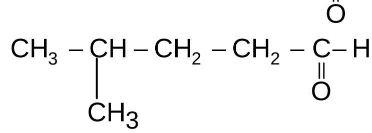
33. 00002.40

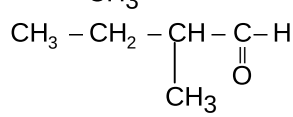
34. 00000.50

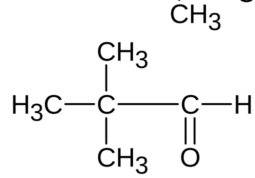
Sol. (for Q.33 and 34)

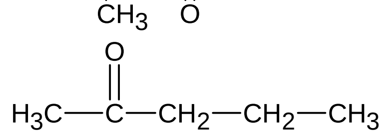
Possible isomers of ‘A ‘ which can produce diastereomeric products.

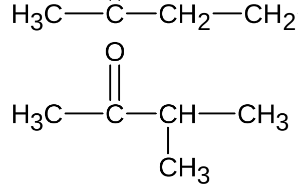
(i) 

(ii) 

(iii) 

(iv) 

(v) 

(vi) 

35. 00002.53

36. 00166.40

Sol. (for Q.35 and 36)

$$\lambda = \frac{0.693}{6.93\times10^{9}} = 10^{-10}\ y^{-1}$$

206 g Pb – 238 g U

20.6 g Pb – 23.8 g U

Thus, during the age of mineral 100 + 23.8 = 123.8 g U disintegrated to 100 g U.

$$t = \frac{2.303}{\lambda}\log\frac{123.8}{100}$$

$$= \frac{2.303}{10^{-10}}\log 1.238$$

$$= 2.303\times0.11\times10^{10}\ y$$

$$= 0.2533\times10^{10}\ y$$

$$= 2.53\times10^{9}\ y$$

$123.8\ g\ U \rightarrow 20.6\ g\ Pb$

$$\therefore 1000\ g\ U \rightarrow \frac{20.6}{123.8}\times1000\ g$$

= 166.397 g Pb

37. 00000.80

38. 00001.53

Sol.

$$\underset{X}{CaCO_3} \xrightarrow{\Delta} \underset{Q}{CaO} + \underset{P}{CO_2}$$

$$CaO + \underset{D}{3C} \xrightarrow{\Delta} \underset{E}{CaC_2} + \underset{K}{CO}$$

$$CaC_2 + 2H_2O \longrightarrow Ca(OH)_2 + \underset{H}{C_2H_2}$$

$$CaC_2 + N_2 \longrightarrow \underset{G}{Ca(NCN)} + \underset{D}{C}$$

$$\underset{G}{Ca(NCN)} + 3H_2O \longrightarrow \underset{I}{2NH_3}\uparrow + \underset{X}{CaCO_3}$$

Mathematics

PART – III

Section – A

39. A

Sol. Let S = $z + 2z^2 + 3z^3 + ... 18z^{18}$

$zS = z^2 + 2z^3 + ... 17z^{18} + 18z^{19}$

$$\Rightarrow S = \frac{z(1-z^{18})}{(1-z)^2} - \frac{18z^{19}}{(1-z)} = -\frac{18(\cos 20° + i\sin 20°)}{(1-\cos 20°) + i\sin 20°}$$

$$\Rightarrow |S| = \frac{18}{2\sin 10°} = \frac{9}{\sin 10°}$$

40. C

Sol. $$\sum_{k=1}^{\infty} \tan^{-1}\left(\frac{4k}{k^4+5}\right) = \sum_{k=1}^{\infty} \tan^{-1}\left(\frac{4k}{1+(k^2+2)^2 - 4k^2}\right)$$

$$= \sum_{k=1}^{\infty}\left(\tan^{-1}(k^2+2k+2) - \tan^{-1}(k^2+2-2k)\right)$$

$$= 2\tan^{-1}(\infty) - \tan^{-1}1 + \tan^{-1}2 = \frac{3\pi}{4} - \tan^{-1}2$$

41. D

Sol. $y^{-1/3}\frac{dy}{dx} + \frac{y^{2/3}}{(x-1)} = x$. Let $y^{2/3} = t$

$$\Rightarrow \frac{dt}{dx} + \frac{2}{3}\frac{t}{(x-1)} = \frac{2x}{3},\ \text{I.F} = (x-1)^{2/3}$$

42. C

Sol. If it is a polynomial, f(x) is of the form ax + b, $f(f(f(x))) = a^3x + a^2b + ab + b$

$\Rightarrow a^3 = 125 \Rightarrow a = 5$ and b = 1

$\therefore f(x) = 5x + 1$

43. B, C

Sol. Let points of hyperbola xy = 16 be $A\left(4t_1, \frac{4}{t_1}\right)$, $B\left(4t_2, \frac{4}{t_2}\right)$ slope of AB = 1 $\Rightarrow t_1t_2 = -1$

Equation of circle $(x - 4t_1)(x - 4t_2) + \left(y - \frac{4}{t_1}\right)\left(y - \frac{4}{t_2}\right) = 0$

$x^2 + y^2 - 32 + (t_1 + t_2)(4y - 4x) = 0$

It is of the form $S + \lambda L = 0$ $\therefore$ it passes through points of intersection of $x^2 + y^2 = 32$ and y = x

44. A, C

Sol. $$T_n = \frac{n^2+2n-1}{(n+3)!} = \frac{(n^2+3n)-(n+1)}{(n+3)!} = \frac{n}{(n+2)!} - \frac{n+1}{(n+3)!}$$

$$S_n = \frac{1}{3!} - \frac{2}{4!} + \frac{2}{4!} - \frac{3}{5!} + \frac{3}{5!} - \frac{4}{6!} + ... \frac{n}{(n+2)!} - \frac{n+1}{(n+3)!} = \frac{1}{3!} - \frac{n+1}{(n+3)!}$$

$$T_{10} = \frac{119}{13!},\ T_{14} = \frac{223}{17!},\ S_{\infty} = \frac{1}{6}$$

45. B, C, D

Sol. x > sin x in $\left[0, \frac{\pi}{2}\right]$ $\therefore$ cos x < cos(sin x) $\Rightarrow I_3 < I_1$ and sin(cos x) < cos x $\Rightarrow I_2 < I_3$

46. B, C

Sol. $\lim\limits_{x\to1^-} f(x) = \log_e 3$; $\lim\limits_{x\to1^+} f(x) = -\log_e 3$; $\lim\limits_{x\to-1^-} f(x) = \lim\limits_{x\to-1^+} f(x) = \log_e 3$

47. A, C

Sol. Any point on line L is (3k + 2, 4k + 1, 5k + 6) $\therefore$ k = 0 $\therefore$ A is (2, 1, 6)

Image $A'(x_2, y_2, z_2)$; $\frac{x_2-2}{3} = \frac{y_2-1}{4} = \frac{z_2-6}{5} = -2\left(\frac{2+1-12-3}{6}\right) = 4$

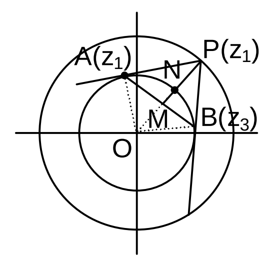
$\therefore$ A′(6, 5, –2) putting (3k + 2, 4k + 1, 5k + 6) in equation of plane, we get B

3k + 2 + 4k + 1 – 10k – 12 = 3 $\Rightarrow$ k = –4 $\therefore$ B is (–10, –15, –14)

$\therefore$ Equation of reflected ray BA′ is $\frac{x+10}{4} = \frac{y+15}{5} = \frac{z+14}{3}$

48. A, B, C, D

Sol. OA = OB = 1, OP = 2



$\sin OPA = \frac{1}{2}$ $\therefore$ $\angle OPA = \frac{\pi}{6}$ and $\angle APB = \frac{\pi}{3}$

OM $\perp$ AB so ΔAPB is equilateral

$OM = \frac{1}{2}$, $MN = \frac{1}{2}$, PN = 1

So centroid lies on |z| = 1

$\left|\frac{z_1+z_2+z_3}{3}\right| = 1 \Rightarrow \left|z_1^2+z_2^2+z_3\right|^2 = 9$

$\Rightarrow (z_1+z_2+z_3)(\bar{z}_1+\bar{z}_2+\bar{z}_3) = 9 \Rightarrow \left(\frac{4}{z_1}+\frac{1}{z_2}+\frac{1}{z_3}\right)\left(\frac{4}{z_1}+\frac{1}{z_2}+\frac{1}{z_3}\right) = 9$

Section – B

49. 3

Sol. Put x = 0, y = 0, we get $f(0) = \frac{1}{2}$

$\lim\limits_{h\to0} \frac{f(0+h)-\frac{1}{2}}{2h} = \frac{f'(0)}{2} = 1 \Rightarrow f'(0) = 2$

Using partial differentiation w.r.t x (and treating y as constant)

$f'(x + y) = f'(x) + 4xy + 2y^2$; put x = 0, y = x $\Rightarrow f'(x) = f'(0) + 2x^2$

$\Rightarrow f'(x) = 2 + 2x^2 \Rightarrow f(x) = 2x + \frac{2x^3}{3} + c$; put x = 0, we get $c = \frac{1}{2}$

$\Rightarrow f(x) = 2x + \frac{2x^3}{3} + \frac{1}{2}$

50. 8

Sol. D = 0 and $a_0 + a_1 + a_2 \neq 0$ $\therefore$ $a_0 = a_1 = a_2$

Put x = 0 $\Rightarrow 1 = a_0 \Rightarrow a_1 = a_2 = 1$

Differentiate both sides w.r.t x

$4(1 + ax + bx^2)^3 \times (a + 2bx) = a_1 + 2a_2x + ... 8a_8x^7$

Put x = 0 $\Rightarrow$ 4a = $a_1 \Rightarrow a = \frac{1}{4}$

Differentiating again and putting x = 0, we get $b = \frac{5}{32} \;\therefore\; \frac{5\times\frac{1}{4}}{\frac{5}{32}} = 8$

51. 1

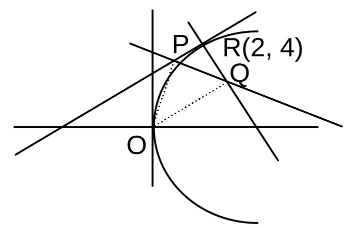
Sol. Equation of tangent at R, x – y + 2 = 0

Equation of normal at P, x + y – 6 = 0

$\therefore$ Combined equation of PR and QR is

$x^2 - y^2 - 4x + 8y - 12 = 0$

Homogenising with the help of kx + y = 3, we get equation of OP and OQ



$$x^2 - y^2 - 4x\left(\frac{kx+y}{3}\right) + 8y\left(\frac{kx+y}{3}\right) - 12\left(\frac{kx+y}{3}\right)^2 = 0$$

Coefficient of x^2 + coefficient of y^2 = 0

$\left(1 - \frac{4k}{3} - \frac{4k^2}{3}\right) + \left(-1 + \frac{8}{3} - \frac{4}{3}\right) = 0$; $k^2 + k - 1 = 0$ $\therefore$ sum of value of k = –1

Section – C

52. 00240.00

53. 00256.00

Sol. (for Q. 52-53)

$$D = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & 3 \\ 1 & 2 & p \end{vmatrix} = -p \;;\; Dx = \begin{vmatrix} 4 & 1 & 1 \\ 6 & 1 & 3 \\ q & 2 & p \end{vmatrix} = -2(p - q + 6);$$

$$Dy = \begin{vmatrix} 1 & 4 & 1 \\ 2 & 6 & 3 \\ 1 & q & p \end{vmatrix} = -2p - q + 6 \;;\; D3 = \begin{vmatrix} 1 & 1 & 4 \\ 2 & 1 & 6 \\ 1 & 2 & q \end{vmatrix} = 6 - q$$

For unique solution $D \neq 0 \Rightarrow p \neq 0$ $\therefore$ total number of ordered pairs = 15 $\times$ 16 = 240 $\therefore$ A = 240

For no solution p = 0, $q \neq 6$ $\therefore$ total number of ordered pairs = 1 $\times$ 15 = 15 $\therefore$ B = 15

For infinite solution p = 0, q = 6 $\therefore$ total number of ordered pairs = 1 $\times$ 1 = 1 $\therefore$ C = 1

54. 00003.00

55. 00002.00

Sol. $f'(x) = 0 \Rightarrow f(x)$ = constant $\because$ f(3) = λ $\therefore$ f(x) = λ

$$\therefore\; \frac{f(1)+f(2)+f(3)}{f(3)} = 3$$

$$P^2 = P\cdot P = \begin{bmatrix} \cos\frac{2\pi}{9} & \sin\frac{2\pi}{9} \\ -\sin\frac{2\pi}{9} & \cos\frac{2\pi}{9} \end{bmatrix} \;;\; P^n = \begin{bmatrix} \cos\frac{n\pi}{9} & \sin\frac{n\pi}{9} \\ -\sin\frac{n\pi}{9} & \cos\frac{n\pi}{9} \end{bmatrix}$$

$$aP^6 + bP^3 + cI = a\begin{bmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix} + b\begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} + c\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$\Rightarrow -\frac{a}{2} + \frac{b}{2} + C = 0 \;;\; \frac{\sqrt{3}}{2}(a+b) = 0 \;\Rightarrow c = a \Rightarrow a = -b$$

56. 00000.33

57. 00000.63

Sol. $f(1) = \left|\sin\left((2p_1 - 1)\frac{\pi}{6}\right)\right| + \left|\cos\left(\frac{p_2\pi}{6}\right)\right|$

Possible values of $(2p_1 - 1)\frac{\pi}{6} = \frac{\pi}{6}, \frac{3\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{9\pi}{6}, \frac{11\pi}{6}, \frac{13\pi}{6}, \frac{15\pi}{6}$

Possible values of $\frac{p_2\pi}{6} = \frac{\pi}{6}, \frac{2\pi}{6}, \frac{3\pi}{6}, \frac{4\pi}{6}, \frac{5\pi}{6}, \frac{6\pi}{6}, \frac{7\pi}{6}, \frac{8\pi}{6}$

For f(1) to be an integer $(2p_1 - 1)\frac{\pi}{6} = \frac{3\pi}{6}, \frac{9\pi}{6}, \frac{15\pi}{6}$ and $\frac{p_2\pi}{6} = \frac{3\pi}{6}, \frac{6\pi}{6}$

or $(2p_1 - 1)\frac{\pi}{6} = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}, \frac{13\pi}{6}$ and $p_2 = \frac{2\pi}{6}, \frac{4\pi}{6}, \frac{8\pi}{6}$

$\therefore$ Required probability $\frac{3}{8} \times \frac{2}{8} + \frac{5}{8} \times \frac{3}{8} = \frac{21}{64} = 0.328$

$f(2) = \left|\sin(2p_1 - 1)\frac{\pi}{3}\right| + \left|\cos\frac{p_2\pi}{3}\right|$ possible values of $(2p_1 - 1)\frac{\pi}{3}$ are

$\frac{\pi}{3}, \frac{3\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{9\pi}{3}, \frac{11\pi}{3}, \frac{13\pi}{3}, \frac{15\pi}{3}$

Possible values of $\frac{p_2\pi}{3}$ are $\frac{\pi}{3}, \frac{2\pi}{3}, \frac{3\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}, \frac{6\pi}{3}, \frac{7\pi}{3}, \frac{8\pi}{3}$ for f(2) to be an irrational number

$(2p_1 - 1)\frac{\pi}{3} = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}, \frac{13\pi}{3}$; $\frac{p_2\pi}{3} = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{3\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}, \frac{6\pi}{3}, \frac{7\pi}{3}, \frac{8\pi}{3}$

$\therefore$ Required probability $\frac{5}{8} \times \frac{8}{8} = \frac{5}{8} = 0.625$